

Exercises on Slutsky's Theorem

10 practice problems with hints (no solutions).

Exercise 1. If $X_n \rightarrow X$ in probability and $Y_n \rightarrow 5$ in probability, prove that $X_n + Y_n \rightarrow X + 5$ in probability.

Hint: Use the triangle inequality and the epsilon/2 argument.

Exercise 2. If $X_n \rightarrow X$ in probability and $Y_n \rightarrow -2$ in probability, prove that $X_n - Y_n \rightarrow X + 2$ in probability.

Hint: Rewrite subtraction as addition.

Exercise 3. If $X_n \rightarrow X$ in probability and $Y_n \rightarrow 3$ in probability, prove that $X_n Y_n \rightarrow 3X$ in probability.

Hint: Write $X_n Y_n - 3X = X_n(Y_n - 3) + 3(X_n - X)$.

Exercise 4. If $X_n \rightarrow X$ in probability and $Y_n \rightarrow c \neq 0$ in probability, prove that $X_n/Y_n - X/c$ converges to 0 in probability.

Hint: Combine fractions and control the event $\{|Y_n| \text{ is small}\}$.

Exercise 5. Suppose $X_n \rightarrow 2$ and $Y_n \rightarrow 4$ in probability. Show that $2X_n - 3Y_n \rightarrow -8$ in probability.

Hint: Express as sums and scalar multiples.

Exercise 6. Suppose $X_n \rightarrow X$ and $Y_n \rightarrow 0$ in probability. Show that $X_n Y_n \rightarrow 0$ in probability.

Hint: Use the multiplication version of Slutsky and view 0 as a constant.

Exercise 7. If $X_n \rightarrow X$ in probability, show that $X_n + 1/n \rightarrow X$ in probability.

Hint: Treat $1/n$ as a deterministic sequence converging to 0.

Exercise 8. If $X_n \rightarrow X$ in probability and $Y_n \rightarrow 1$ in probability, show that $X_n/(1+Y_n) \rightarrow X/2$ in probability.

Hint: First show $1+Y_n \rightarrow 2$, then apply the division form.

Exercise 9. Let $X_n \rightarrow X$ and $Y_n \rightarrow c$ in probability. Show that $(X_n + Y_n)(X_n - Y_n) \rightarrow (X+c)(X-c)$ in probability.

Hint: Apply addition/subtraction first, then multiplication.

Exercise 10. Suppose $X_n \rightarrow X$ and $Y_n \rightarrow c \neq 0$ in probability. Show that $(X_n^2 + Y_n)/(Y_n) \rightarrow (X^2 + c)/c$ in probability.

Hint: Break the expression into simpler pieces and repeatedly apply Slutsky's theorem.